SAHAD MUHAMMED VILLAN

Freiburg, Germany | +49 176 37025953 | sahadmhd1@gmail.com | linkedin.com/in/sahadmuhammedv | github.com/sahadvillan

PROFESSIONAL SUMMARY

Machine Learning Engineer specialized in building production-ready AI systems for medical imaging, computer-assisted surgery, and diagnostic hardware. Experienced in PyTorch pipelines, real-time spatial tracking, explainable AI (SHAP), and uncertainty estimation. Proven ability to deploy low-latency inference services using Docker and FastAPI, bridging medical hardware architectures with deep learning for robust clinical software applications.

TECHNICAL SKILLS

Surgical AI & ML	PyTorch, Computer Vision, Few-Shot Learning, Capsule Networks, Explainable AI (SHAP), Uncertainty Estimation
Medical Imaging & 3D	Multispectral / Hyperspectral (HSI), MRI (DICOM), Image Registration, 3D Reconstruction, clinical visualization
MLOps & Software	C++, Python, OpenCV, ITK, FastAPI, Docker, CI/CD (GitHub Actions), SQL, Linux, Git
Systems & Hardware	Sensor calibration, optical tracking (Polaris), medical hardware integration, real-time data acquisition, validation protocols
Languages	English (Fluent), German (Intermediate / Professional)

PROFESSIONAL EXPERIENCE

Master Thesis Student (R&D) – Medical Imaging & AI *May 2025 – Oct 2025*

SCHÖLLY FIBEROPTIC GMBH, Denzlingen, Germany

- Designed and implemented a modular PyTorch pipeline for tissue oxygen saturation (StO_2) estimation from multispectral data, reducing Mean Absolute Error (MAE) by **25–30%**.
- Integrated Monte Carlo Dropout uncertainty estimation (**>92% calibrated coverage**) and SHAP explainability to support confidence-aware medical image interpretation.
- Deployed reproducible GPU training workflows using Docker, Azure ML, and built a clinician-facing GUI for real-time prediction visualization.

Project Intern – Medical Image Processing *May 2024 – Oct 2024*

ICCAS – Innovation Center Computer Assisted Surgery, Leipzig, Germany

- Developed spectral-spatial deep learning models (Capsule Networks) in PyTorch for hyperspectral medical image segmentation in computer-assisted surgery setups.
- Benchmarked advanced architectures to ensure boundary and segmentation robustness under noise, camera movement, and variable illumination.
- Automated training and evaluation workflows using GitHub Actions CI/CD to ensure reproducible and traceable research results.

Research & Development Engineer *Jul 2022 – Jul 2023*

Ubio BioTechnology Systems Pvt Ltd, Kochi, India

- Designed OpenCV-based computer vision pipelines to extract quantitative metrics from raw sensor data, adhering to structured medical device design documentation standards.
- Developed high-performance, low-latency FastAPI inference services containerized with Docker for Point-of-Care diagnostic assay readers.

Engineering & Research Engineer (Intern) *Mar 2020 – May 2022*

Ubio BioTechnology Systems Pvt Ltd, Kochi, India

- Developed early-stage image processing prototypes using OpenCV for automatic feature localization.
- Assisted in designing sensor calibration routines and contributed to technical software verification and documentation lifecycle processes.

SELECTED PROJECTS

Surgical Tissue Localization & Deformation Tracking

- Developed a PyTorch-based anatomical landmark localization and tissue deformation tracking pipeline using **CoTracker3**.
- Implemented **Monte Carlo (MC) Dropout** to quantify spatial tracking uncertainty (reliability halos) for clinical decision safety.
- Modeled local tissue flow (motion fields) and strain metrics (compression/stretching) during active tissue deformation.

Brain Tumor Detection & Segmentation (MRI)

- Developed ResNet/ResUNet classification and segmentation models on 7,800 MRI scans, achieving **92%** accuracy.

EDUCATION

M.Sc. Biomedical Engineering *2023 – 2025*

Anhalt University of Applied Sciences, Germany

B.Tech Instrumentation Technology *2018 – 2022*

Cochin University of Science and Technology, India